

Mercor Problem - Felipe Maia Silva

[0] Problem 1. In this exercise, we will study properties of the eccentricity of orbits and understand how it relates to energy and angular momentum. Denoting $\mu = GM$ and $h = L/m$, do what is asked in the following items.

Part I: The Eccentricity Vector

- a) Write an expression for Newton's second law in vector format for a body of mass m orbiting a body of mass M at a distance r .
- b) Take the cross product of your expression with $\mathbf{h}$ and prove that:

$$\frac{d}{dt}(\dot{\mathbf{r}} \times \mathbf{h}) = \frac{\mu}{r} \mathbf{v} - \frac{\mu \dot{r}}{r^2} \mathbf{r}$$

Where $\dot{\mathbf{r}}$ represents the radial velocity and $\mathbf{v}$ the velocity.

- c) We can write the right side of the previous equation as:

$$\frac{d}{dt}(A\mathbf{r})$$

Find the value of A .

- d) Integrate both sides of the found equality and take the scalar product of both sides with $\mathbf{r}$. After that, solve for r . Your result should be very similar to what is predicted only by geometry:

$$r = \frac{p}{1 + e \cos \theta}$$

Find the values of p and e . **Don't forget the integration constant. Your answers may depend on it.**

- e) In the previous item, e is the eccentricity of the orbit. Returning to the equation initially obtained in item c), after integration, find a way to express $\mathbf{e}$, as a vector with magnitude e that points directly to the orbited body.
- f) Prove that the relationship found in the previous item can be written as:

$$\mu \mathbf{e} = (v^2 - \frac{\mu}{r}) \mathbf{r} - (\mathbf{r} \cdot \mathbf{v}) \mathbf{v}$$

Where does this vector point?

Part II: Determining Orbital Elements from the Eccentricity Vector

Let's define our coordinate system in the following manner. Consider a right-handed system, oriented such that the xy plane coincides with the equatorial plane and with $\mathbf{z}$ pointing to the North pole. Defining the nodal vector as $\mathbf{n} = \mathbf{z} \times \mathbf{h}$.

- a) Find an expression for $\mathbf{h}$ as a function of the components of the vector $\mathbf{r}$ and the vector $\mathbf{v}$. After that, find the vector $\mathbf{n}$ as a function of the components of $\mathbf{h}$.

- b) Draw a celestial sphere and represent on it: the equatorial plane and an arbitrary orbit (with an inclination different from 0). On it, mark the following angles: i , the orbital inclination, Ω the longitude of the ascending node, ω the argument of periapsis, and θ , the true anomaly.
- c) Find expressions for all these angles as a function of $\mathbf{e}$, $\mathbf{h}$, $\mathbf{n}$, $\mathbf{r}$ and any of the vectors defined in our xyz coordinate system.

Solution

Part I

- a) Using the vector form of the gravitational force, we have:

$$-\frac{GMm}{r^3}\mathbf{r} = m\ddot{\mathbf{r}} \rightarrow \boxed{\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r}}$$

- b) Taking the cross product of this expression with $\mathbf{h}$, we have:

$$\ddot{\mathbf{r}} \times \mathbf{h} = -\frac{\mu}{r^3}(\mathbf{r} \times \mathbf{h}) = \frac{\mu}{r^3}(\mathbf{h} \times \mathbf{r})$$

Since $\mathbf{h}$ is constant, the left side of the equation can be written as $d(\dot{\mathbf{r}} \times \mathbf{h})/dt$. For the right side, let's write $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ and use the BAC-CAB rule, which states that $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$. Thus:

$$\frac{d}{dt}(\dot{\mathbf{r}} \times \mathbf{h}) = \frac{\mu}{r^3}((\mathbf{r} \times \mathbf{v}) \times \mathbf{r}) = \frac{\mu}{r^3}(r^2\mathbf{v} - \mathbf{r}(\mathbf{r} \cdot \mathbf{v}))$$

Note that $\mathbf{r} \cdot \mathbf{v}$ is the multiplication of the components of $\mathbf{r}$ and $\mathbf{v}$ that are in the same direction. Thus, $\mathbf{r} \cdot \mathbf{v} = r\dot{r}$. Substituting into the previous expression:

$$\boxed{\frac{d}{dt}(\dot{\mathbf{r}} \times \mathbf{h}) = \frac{\mu}{r}\mathbf{v} - \frac{\mu\dot{r}}{r^2}\mathbf{r}}$$

- c) Expanding the expression provided in the statement:

$$\frac{d}{dt}(A\mathbf{r}) = \mathbf{r}\frac{d}{dt}A + A\mathbf{v}$$

Comparing the expressions, it is easy to see that $\boxed{A = \mu/r}$.

- d) Using the information obtained in the previous items, we have:

$$\frac{d}{dt}(\dot{\mathbf{r}} \times \mathbf{h}) = \mu \frac{d}{dt} \left(\frac{\mathbf{r}}{r} \right)$$

Multiplying both sides by dt and integrating, we have:

$$\dot{\mathbf{r}} \times \mathbf{h} = \frac{\mu}{r}\mathbf{r} + \mathbf{B}$$

Where $\mathbf{B}$ is a constant integration vector. Now, taking the scalar product with $\mathbf{r}$:

$$\mathbf{r} \cdot (\dot{\mathbf{r}} \times \mathbf{h}) = \mu r + \mathbf{r} \cdot \mathbf{B}$$

Using $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$ and defining ν as the angle between $\mathbf{r}$ and $\mathbf{B}$, we have:

$$h^2 = \mu r + r B \cos \nu$$

Solving for r , we arrive at:

$$r = \frac{h^2/\mu}{1 + (B/\mu) \cos \nu}$$

Thus:

$$p = \frac{h^2}{\mu}, \quad e = \frac{B}{\mu}$$

e) Using $\mathbf{e} = \mathbf{B}/\mu$ and returning to item d), we have:

$$\dot{\mathbf{r}} \times \mathbf{h} = \frac{\mu}{r} \mathbf{r} + \mu \mathbf{e}$$

Solving for $\mathbf{e}$ and noting that $\dot{\mathbf{r}} \times \mathbf{h} = \mathbf{v} \times \mathbf{h}$, we have:

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}$$

f) Writing $\mathbf{h} = \mathbf{r} \times \mathbf{v}$, we have:

$$\mathbf{e} = \frac{\mathbf{v} \times (\mathbf{r} \times \mathbf{v})}{\mu} - \frac{\mathbf{r}}{r}$$

Multiplying both sides of the expression by μ , and using BAC-CAB, we have:

$$\mu \mathbf{e} = v^2 \mathbf{r} - \mathbf{v}(\mathbf{v} \cdot \mathbf{r}) - \frac{\mu}{r} \mathbf{r}$$

Reorganizing the terms:

$$\mu \mathbf{e} = (v^2 - \frac{\mu}{r}) \mathbf{r} - (\mathbf{r} \cdot \mathbf{v}) \mathbf{v}$$

Thus, this vector points towards the *periapsis*.

Part II: Determining Orbital Elements from the Eccentricity Vector

- a) The vector $\mathbf{h}$ is defined by $\mathbf{h} = \mathbf{r} \times \mathbf{v}$. By definition,

$$\mathbf{h} = \mathbf{r} \times \mathbf{v} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ r_x & r_y & r_z \\ v_x & v_y & v_z \end{vmatrix} = (r_y v_z - r_z v_y) \hat{x} + (r_z v_x - r_x v_z) \hat{y} + (r_x v_y - r_y v_z) \hat{z}$$

Similarly, $\mathbf{n} = \mathbf{z} \times \mathbf{h}$

$$\mathbf{n} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & 1 \\ h_x & h_y & h_z \end{vmatrix} = -h_y \hat{x} + h_x \hat{y}$$

- b) The drawing should resemble the following figure:

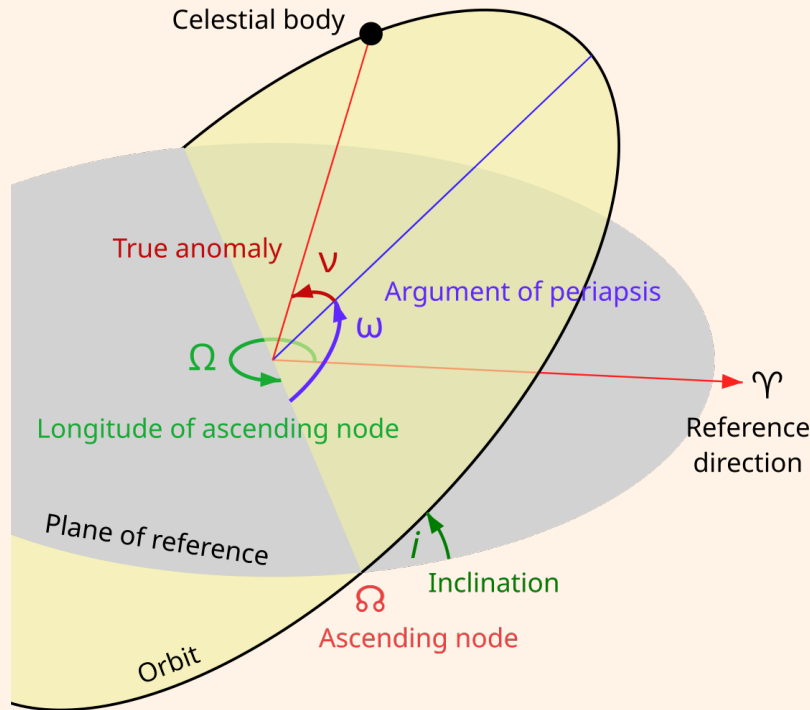


Figure 1: Source Wikipedia

- c) To orient ourselves, let's create a coordinate system where the North pole corresponds to the z -axis and the x -axis points in the direction of the vernal point. Starting with the inclination, it is the angle between the North pole and the perpendicular plane of the orbit. Thus:

$$\mathbf{z} \cdot \mathbf{h} = \cos i |\mathbf{z}| |\mathbf{h}| \rightarrow i = \cos^{-1} \left(\frac{\mathbf{z} \cdot \mathbf{h}}{|\mathbf{z}| |\mathbf{h}|} \right)$$

Using similar arguments, we can obtain:

$$\Omega = \cos^{-1} \left(\frac{\mathbf{x} \cdot \mathbf{n}}{|\mathbf{x}| |\mathbf{n}|} \right)$$

$$\omega = \cos^{-1} \left(\frac{\mathbf{n} \cdot \mathbf{e}}{|\mathbf{n}| |\mathbf{e}|} \right)$$

$$\theta = \cos^{-1} \left(\frac{\mathbf{e} \cdot \mathbf{r}}{|\mathbf{e}| |\mathbf{r}|} \right)$$